

TITLE:

**AI COMIC STORY
CREATOR USING
GEMINI MODELS**

Abstract :

The AI Comic Story Creator Using Gemini Models is an intelligent content-generation system that helps users create comic stories from simple text prompts.

Traditional comic creation requires separate skills for story writing, character design, scene planning, dialogue writing, and visual composition.

This project combines these stages into a single AI-assisted workflow.

Gemini models are used to understand the user's prompt and generate a structured story containing a title, characters, scenes, narration, and dialogue.

The generated content can be converted into a panel-by-panel storyboard and, where image generation is integrated, illustrated comic panels.

The system aims to provide a simple, creative, and accessible platform for students, writers, educators, and hobbyists.

The project also demonstrates practical concepts such as prompt engineering, API integration, structured AI output, user input validation, error handling, and responsible use of generative AI

Introduction :

Artificial Intelligence has changed the way digital content is created.

Generative AI systems can understand natural-language instructions and produce text, images, code, and other creative content.

Comic creation is a suitable application because it combines narrative writing and visual storytelling.

This project applies Gemini models to the comic-creation process.

Instead of writing every scene manually, the user can enter an idea such as a fantasy adventure, educational story, science-fiction episode, or children's story.

The AI analyzes the idea and creates a structured comic plan.

The application can be implemented as a web application with a simple user interface.

The frontend collects the user's requirements, the backend communicates with the Gemini API, and the generated response is formatted into comic panels.

Additional image-generation capabilities can be connected when required.

Problem Statement:

Objectives Generate original comic story ideas from user prompts.

Create titles, characters, settings, scenes, narration, and dialogue.

Divide a story into clear comic panels.

Use Gemini models for natural-language understanding and generation.

Provide a simple interface for entering and editing story requirements.

Reduce the time required for initial comic story development.

Support different genres, tones, languages, and story lengths. Demonstrate practical integration of a generative-AI API into an application.

Scope of the Project:

The system is primarily focused on AI-assisted comic story planning and generation.

It can be extended from a text-only story generator to a complete comic production tool.

- Story idea generation
- Character profile generation
- Scene and panel planning
- Dialogue and narration generation
- Comic-panel prompt generation
- Optional image generation
- Story editing and regeneration
- Export to PDF or image format
- Saving and managing comic projects

Main Modules:

- **User Input Module:** Accepts topic, genre, characters, setting, tone, language, and number of panels.
- **Prompt Engineering Module:** Converts user requirements into a clear instruction for Gemini
- **Story Generation Module:** Generates title, plot, characters, narration, and dialogue.
- **Story board Module:** Converts the story into ordered comic panels.
- **Image Prompt Module:** Creates visual descriptions for optional image generation.
- **Comic Layout Module:** Combines panels, narration, and dialogue into pages.
- **ExportModule:** Exports the final result as images, PDF, or other supported formats.
- **Project Management Module:** Optionally saves, loads, edits, and deletes comic projects.

Gemini Model Integration:

Gemini is used as the core language-generation component.

The application sends a prompt containing the user's requirements and requests a structured response.

A recommended response structure contains fields such as title, genre, characters, setting, panels, narration, dialogue, and visual prompt.

Example logical request:

User prompt → System instructions → Gemini model → Structured JSON/story response → Application parser → Comic panels

For reliable application behavior, prompts should explicitly request the required fields and formatting. The application should still validate the returned content rather than assuming that every AI response is perfectly structured.

User Interface:

A typical interface contains:

- Project title field
- Story idea text area
- Genre selection
- Character details
- Language selection
- Number of panels
- Generate Story button
- Regenerate/Edit controls
- Panel preview area
- Export button

How to Run the Project:

- Open a terminal in the project directory.
- Activate the virtual environment if using Python.
- Set the Gemini API key as an environment variable.
- Install dependencies from requirements.txt.
- Start the application using the project's main command.
- Open the local URL displayed by the framework.
- Enter a story idea and generate the comic.
- Save or export the generated result.

Advantages:

- Reduces the time required to create an initial comic story.
- Allows users without advanced writing skills to create structured stories.
- Supports creative experimentation with genres and characters.
- Provides a single workflow for story and storyboard generation.

- Can be extended with image generation and export features.
- Demonstrates real-world use of generative AI.

Future Enhancements:

- Automatic character reference images and consistent character appearance.
- Advanced comic page layouts and speech-bubble placement.
- Voice narration and text-to-speech.
- Multilingual comic generation.
- Automatic PDF/e-book creation.
- Cloud project storage and user accounts.
- Fine-grained editing of individual panels.
- More model choices and configurable AI providers.
- Educational mode for creating comics from academic topics.
- Accessibility features such as alt text and narration.

Conclusion:

The AI Comic Story Creator Using Gemini Models demonstrates how generative AI can be integrated into a practical creative application.

By combining natural-language prompting, Gemini-based story generation, structured storyboard creation, and optional image generation, the project provides an efficient workflow for producing comic concepts.

The project is also a useful demonstration of modern software development because it combines frontend design, backend programming, API integration, prompt engineering, data handling, and version control.

With further development, the system can become a complete platform for creating, editing, saving, and publishing AI-assisted comics.

References:

Google Gemini / Google AI developer documentation – <https://ai.google.dev/>

Google Generative AI documentation – <https://ai.google.dev/gemini-api/docs>

GitHub documentation – <https://docs.github.com/>

Python documentation – <https://docs.python.org/>

MDN Web Docs – <https://developer.mozilla.org/>